

Energy Assessment Test

studyclix

Name:

Oisín Argand

$\frac{77}{80}$ 96%

Well Done
Oisín on an
excellent
result.



EDW

Oct 25

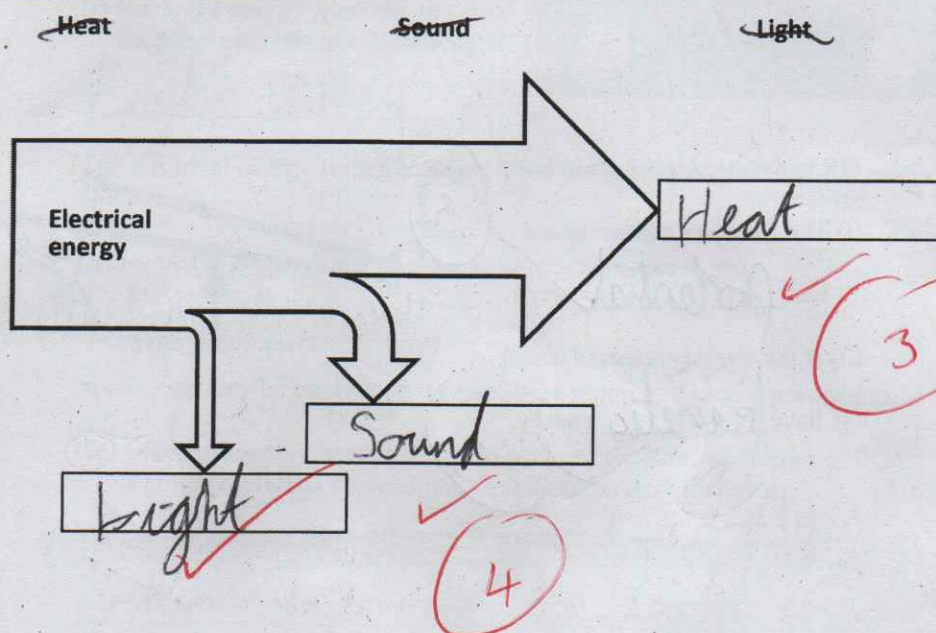
Oisín Argand
4/11/25

Question 1

Electrical appliances use electrical energy to perform a specific function.
The electric kettle in the picture has an electrical power of 2300 W.

Electrical energy is converted to other forms of energy when a kettle is in use.
The kettle in the image above is energy efficient.

- (d) Three forms of energy are listed below.
Use these words to label the Sankey diagram for the kettle.



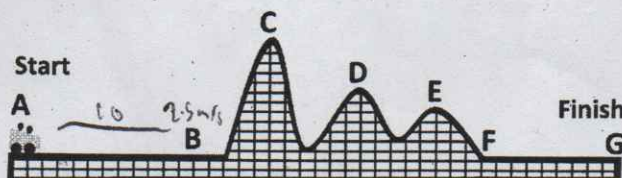
Question 2

Question 11

(30 marks)

The diagram shows a rollercoaster car at the start of its track. The letters A to G indicate the position of the car at certain stages during its journey along the track.

The car is stationary (not moving) at point A. It takes 10 seconds for the car to reach point B. As the car passes point B it is travelling at a speed of 2.5 m/s.



- (c) Which letter represents the point on the track when the car has the most potential energy?
Justify your answer.

C because the drop after it is the highest and steepest meaning there is more potential energy.

6 6

Question 3

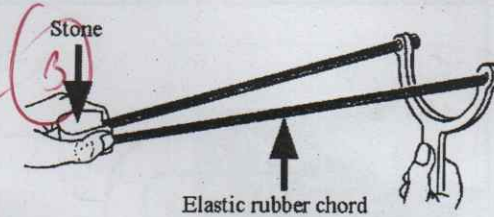
- (b) There are many common forms of energy which can be converted from one form to another in a mobile phone. (9)
- (i) State the standard unit of energy. Joules ✓ (2)
- (ii) When the battery of a mobile phone is being charged, electrical energy is converted to which other form of energy? Chemical energy (3)
- (iii) When a mobile phone is used to listen to music, the energy in the battery is then converted to which other form of energy? Sound energy (3)

Question 4

- (a) Fill in the missing words in both sentences.

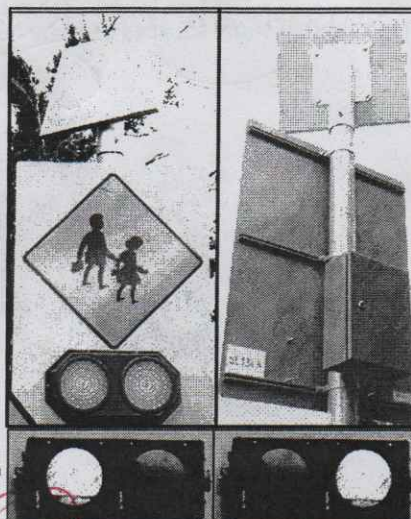
(i) The stretched rubber chord has potential energy.

(ii) If the stone is released it will have kinetic energy.



Question 5

- (b) The top two photographs show the front (left) and the rear (right) of a sign warning motorists approaching a school to take care. The photograph underneath shows the *lights flashing* amber alternately. This happens when the *pupils are coming to school* and are *going home from school*. The *rectangular panel* above the sign is a *solar (photovoltaic) panel*. It *changes energy from the sun* into *electrical energy*.



- (i) Name the *energy from the sun* that the panel changes into electricity. (3)

Or solar

Light energy

The electrical energy is then changed into a *form of energy* that can be *stored* in a *battery*.

- (ii) Name the *form of energy* that can be stored in a *battery*. (3)

Chemical energy

In *winter* it may be *dark* when the pupils are going to or coming from school.

- (iii) Give *two energy conversions* that occur to produce the flashes of light warning motorists approaching the school on dark mornings. (6)

lost 3 here

~~Light energy to chemical energy~~
~~Chemical energy to light energy~~
 Chemical energy to heat energy
 Chemical energy to electrical energy

use only

(1) (2)

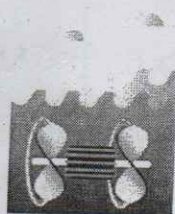
(3)

Question 6

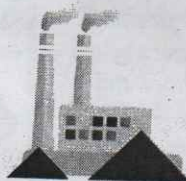
Question 2

(15 marks)

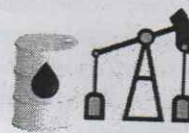
The images below show sources used by humans to generate electricity.



Water



Coal



Oil



Wind



Sun

- (a) Classify each of the sources as renewable or non-renewable by putting a tick (✓) in the correct column.

Source	Renewable	Non-renewable
Water	✓	
Coal		
Oil		✓
Wind	✓	
Sun	✓	

- (b) "Energy producers face ethical issues in the way energy is extracted and/or produced." Do you agree with this statement? Justify your answer.

Yes because large machines drilling into the ground for coal and oil make loud sounds that can disturb the wildlife and local ecosystem

10

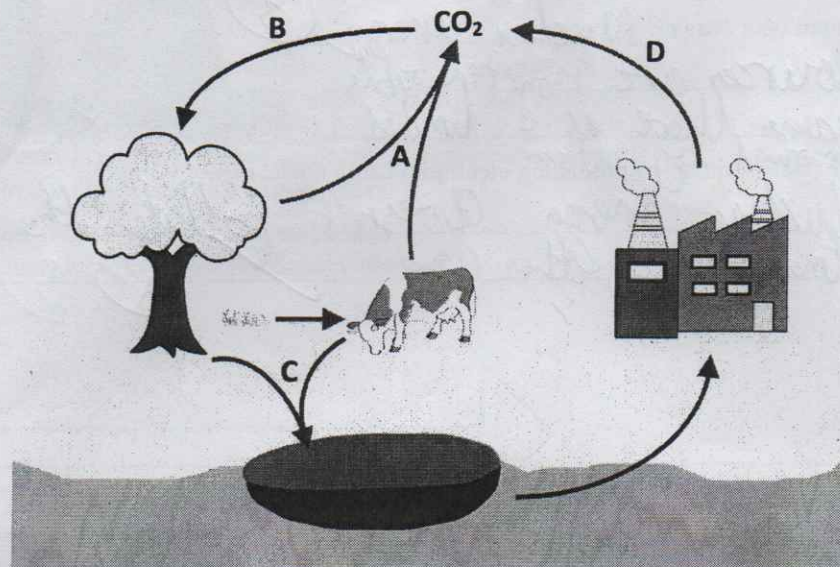
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Question 7

Question 11

(30 marks)

The diagram below shows how carbon moves into and out of the air and soil. This is part of the carbon cycle.



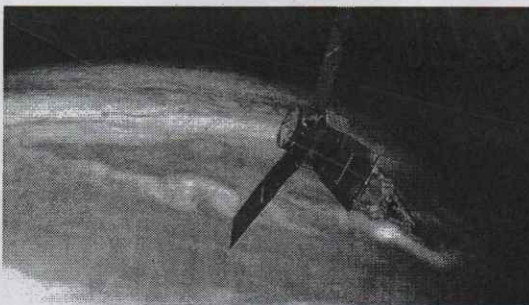
- (d) Process D shows the burning of fossil fuels, which releases carbon dioxide into the air. Name one such fuel.

Coal

✓ (2)

Question 8

(e)



The Sun can also provide power for modern spacecraft. The Juno mission to Jupiter uses solar energy to produce electricity.

Previous long-distance space missions used nuclear power to produce electricity.

State one advantage of using solar energy rather than nuclear energy during space exploration.

Solar energy is a renewable source of energy, so the exploration can continue to get energy from the sun forever

✓ (3)

Question 9

- (h) France produces most of its electricity using nuclear sources, such as uranium.
- (i) Are nuclear energy sources considered to be renewable or non-renewable sources of energy?

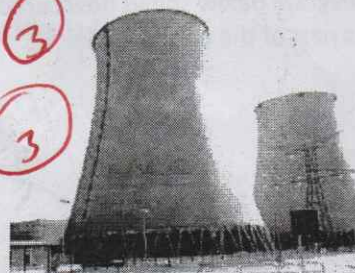
Non-renewable

Explain your answer. Nuclear energy

sources are non-renewable because there is a limited amount of them

- (ii) State one advantage of producing electricity using nuclear energy sources.

Nuclear energy doesn't pollute the atmosphere with CO₂



Question 10

- (h) Renewable energies are shown in the picture.

Pick any two of the energies shown in the picture and name your selection.

Energy one

Wind

Energy two

Solar

- (i) Give one advantage associated with each energy you've selected.

Two **different** reasons must be given.

Energy one

Good for windy places

Energy two

Easy to install solar panels

- (ii) Give one disadvantage associated with each energy you've selected.

Two **different** reasons must be given.

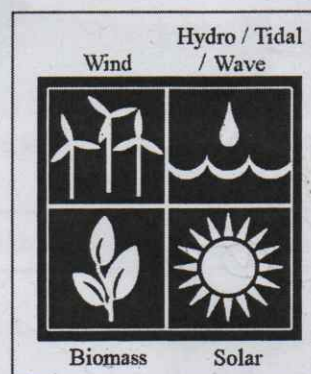
Energy one

Loud

Noise pollution

Energy two

Expensive



Great to see such good use of highlighter!